DFRWS POSTER DRAFT V2

When AI Image Detectors Travel

Source-heldout diagnostics for physical, neural, and frozen-encoder forensics

June 13, 2026

Key Evidence

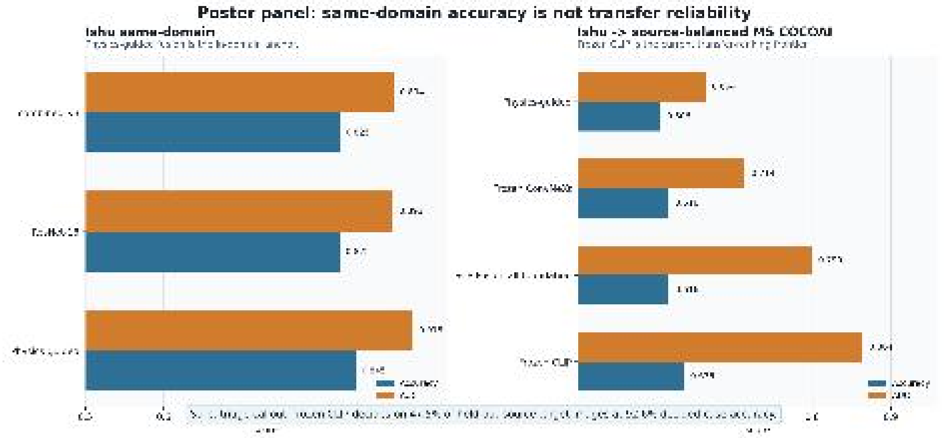
0.845 / 0.918
Physics-guided Ishu
same-domain accuracy / AUC

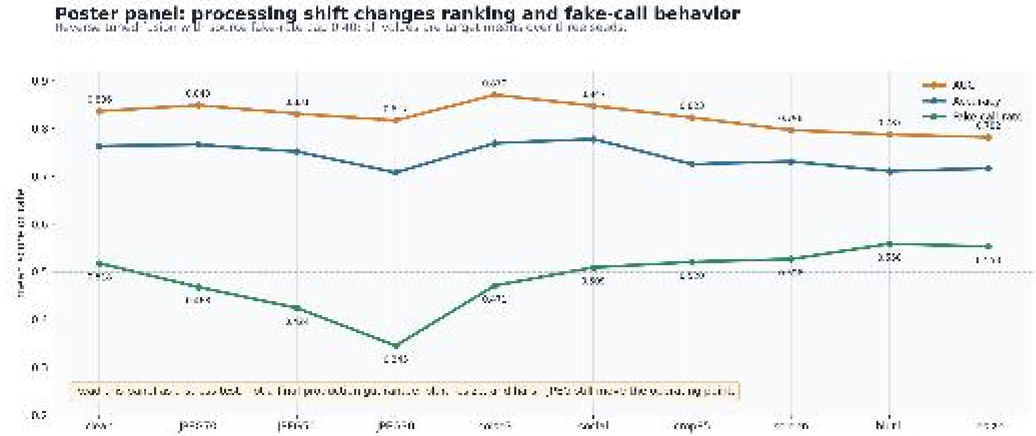
0.636 / 0.864
Frozen CLIP transfer
Ishu -> MS COCOAI accuracy / AUC

47.5% @ 92.6%
Strict CLIP triage
coverage / decided-case accuracy

0.763 / 0.836
Reverse tuned fusion
best capped operating point

Working claim: same-domain wins do not guarantee transfer reliability; ranking, calibration, fake-call rate, and triage split under source shift.





What This Means

- **CLIP is the transfer frontier**
  The current strongest standalone cross-domain ranker is frozen CLIP, not score fusion.
- **Physics helps the anchor**
  Physics-guided ResNet improves the same-domain Ishu result and remains interpretable.
- **Triage is the forensic story**
  High-confidence two-threshold decisions are cleaner than forcing every image through 0.5.
- **Do not overclaim robustness**
  JPEG30, blur, resize, and screenshots still move the operating point.

Next Paper-Critical Step

Run the full repeated-seed combined_v4 transfer check, then decide whether v4 joins the main method or stays an ablation.

Sources: reports/assets/publication_core_results.csv; reports/dfrws_poster_native_figures_2026_06_13.md; public repo github.com/DrStrangel0ve/ai-image-forensics-comparison

Caveat: single-image photometric proxy, not true multi-light photometric stereo.